Exercise 5.1

1. $y(n) = \cos(\omega_0 nT) = \dfrac{e^{j\omega_0 nT} + e^{-j\omega_0 nT}}{2}$

$$Y(z) = Z[y(n)] = \frac{1}{2}\sum_{n=0}^{\infty}\left(e^{j\omega_0 nT} + e^{-j\omega_0 nT}\right)\cdot z^{-n}$$

$$= \frac{1}{2}\left(\sum_{n=0}^{\infty} e^{j\omega_0 nT}\cdot z^{-n} + \sum_{n=0}^{\infty} e^{-j\omega_0 nT}\cdot z^{-n}\right)$$

ZTS:
$$\sum_{n=0}^{\infty} e^{s_0 nT} z^{-n} = \frac{z}{z - e^{s_0 T}}$$

$$= \frac{1}{2}\left(\frac{z}{z - e^{j\omega_0 T}} + \frac{z}{z - e^{-j\omega_0 T}}\right)$$

$$= \frac{1}{2}\,\frac{z(z - e^{-j\omega_0 T}) + z(z - e^{j\omega_0 T})}{(z - e^{j\omega_0 T})(z - e^{-j\omega_0 T})}$$

$$= \frac{z^2 - (\cos\omega_0 T)\,z}{z^2 - 2z\cos(\omega_0 T) + 1}$$

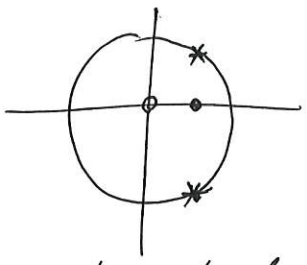
ZTZ

2. $\omega_0 T = 1$

$$Y(z) = \frac{z^2 - 0.54z}{z^2 - 1.08z + 1} = \frac{1 - 0.54z^{-1}}{1 - 1.08z^{-1} + z^{-2}}$$

Zeros: $z_1 = 0$ $z_2 = 0.54$.

poles: $p_1 = 0.54 - 0.8417j$ $p_2 = 0.54 + 0.8417j$



3. Assume sampling interval $t_s = 1$

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sys = tf([1 -0.54 0],[1, -1.08, 1], ts)
bode(sys).
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Exercise 5.2

1. $Y(z)\cdot\left(1-\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}\right)=X(z)\cdot\left(0-z^{-1}+5z^{-2}\right)$

$$H(z)=\frac{Y(z)}{X(z)}=\frac{-z^{-1}+5z^{-2}}{1-\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}}$$

2. ZT 2: when $x(n)$ is unit step $u(n)$, $X(z)=\frac{z}{z-1}$

$$Y(z)=H(z)\cdot X(z)=\frac{-z^{-1}+5z^{-2}}{1-\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}}\cdot\frac{z}{z-1}$$

$$=\frac{-z^{-1}+5z^{-2}}{1-\frac{5}{4}z^{-1}+\frac{1}{8}z^{-2}+\frac{1}{8}z^{-3}}$$

3. To get partial fraction of the above transfer function, you can use 'residue' in Matlab. [r. p. k] = residue([-1, 5 ###], [1 -5/4 1/8 1/8])

$$Y(z)=\frac{-z^2+5z}{z^3-5/4z^2+1/8z+1/8}=$$ ~~~~~~~~~~

$$\frac{Y(z)}{z}=\frac{-z+5}{z^3-5/4z^2+1/8z+1/8}=\frac{6.4}{z-1}+\frac{-12}{z-0.5}+\frac{5.6}{z+0.25}$$

$$Y(z)=6.4\cdot\frac{z}{z-1}-12\cdot\frac{z}{z-0.5}+5.6\cdot\frac{z}{z+0.25}$$

$$y(n)=6.4\cdot 1^n-12\cdot(0.5)^n+5.6\cdot(-0.25)^n.$$

$$=6.4-12e^{-0.6931n}+5.6\,e^{(-1.386+3.142j)n}.$$